UNIT-5

Applications of AI

Natural Language processing

Natural Language Processing (NLP) is a field of Artificial Intelligence (AI) and Computer Science that is concerned with the interactions between computers and humans in natural language. The goal of NLP is to develop algorithms and models that enable computers to understand, interpret, generate, and manipulate human languages.

Common Natural Language Processing (NLP) Task:

- **Text and speech processing:** This includes Speech recognition, text-&-speech processing, encoding(i.e converting speech or text to machine-readable language), etc.
- **Text classification:** This includes Sentiment Analysis in which the machine can analyze the qualities, emotions, and sarcasm from text and also classify it accordingly.
- **Language generation:** This includes tasks such as machine translation, summary writing, essay writing, etc. which aim to produce coherent and fluent text.
- **Language interaction:** This includes tasks such as dialogue systems, voice assistants, and chatbots, which aim to enable natural communication between humans and computers.

The field is divided into three different parts:

1. Speech Recognition—The translation of spoken language into text.
2. Natural Language Understanding (NLU) —The computer's ability to understand what we say.
3. Natural Language Generation (NLG) —The generation of natural language by a computer.

Speech Recognition:

- First, the computer must take natural language and convert it into machine-readable language. This is what speech recognition or speech-to-text does. This is the first step of NLU.
- Hidden Markov Models (HMMs) are used in the majority of voice recognition systems nowadays. These are statistical models that use mathematical calculations to determine what you said in order to convert your speech to text.
- HMMs do this by listening to you talk, breaking it down into small units (typically 10-20 milliseconds), and comparing it to pre-recorded speech to figure out which phoneme you uttered in each unit (a phoneme is the smallest unit of speech). The program then examines the sequence of phonemes and uses statistical analysis to determine the most likely words and sentences you were speaking.

Natural Language Understanding (NLU):

The next and hardest step of NLP is the understanding part.

- First, the computer must comprehend the meaning of each word. It tries to figure out whether the word is a noun or a verb, whether it's in the past or present tense, and so on. This is called Part-of-Speech tagging (POS).
- A lexicon (a vocabulary) and a set of grammatical rules are also built into NLP systems. The most difficult part of NLP is understanding.
- The machine should be able to grasp what you said by the conclusion of the process. There are several challenges in accomplishing this when considering problems such as words having several meanings (polysemy) or different words having similar meanings (synonymy), but developers encode rules into their NLU systems and train them to learn to apply the rules correctly.

Natural Language Generation (NLG):

Future Scope:

- **Bots:** Chatbots assist clients to get to the point quickly by answering inquiries and referring them to relevant resources and products at any time of day or night. To be effective, chatbots must be fast, smart, and easy to use, To accomplish this, chatbots employ NLP to understand language, usually over text or voice-recognition interactions
- **Supporting Invisible UI:** Almost every connection we have with machines involves human communication, both spoken and written. Amazon's Echo is only one illustration of the trend toward putting humans in closer contact with technology in the future. The concept of an invisible or zero user interface will rely on direct communication between the user and the machine, whether by voice, text, or a combination of the two. NLP helps to make this concept a real-world thing.
- **Smarter Search:** NLP's future also includes improved search, something we've been discussing at Expert System for a long time. Smarter search allows a chatbot to understand a customer's request can enable "search like you talk" functionality (much like you could query Siri) rather than focusing on keywords or topics. Google recently announced that NLP capabilities have been added to Google Drive, allowing users to search for documents and content using natural language.

Speech recognition

Speech recognition, or speech-to-text, is the ability of a machine or program to identify words spoken aloud and convert them into readable text. Rudimentary speech recognition software has a limited vocabulary and may only identify words and phrases when spoken clearly. More sophisticated software can handle natural speech, different accents and various languages.

Speech recognition uses a broad array of research in computer science, linguistics and computer engineering. Many modern devices and text-focused programs have speech recognition functions in them to allow for easier or hands-free use of a device.

Speech recognition and voice recognition are two different technologies and should not be confused:

- **Speech recognition** is used to identify words in spoken language.
- **Voice recognition** is a biometric technology for identifying an individual's voice.

How does speech recognition work?

Speech recognition systems use computer algorithms to process and interpret spoken words and convert them into text. A software program turns the sound a microphone records into written language that computers and humans can understand, following these four steps:

1. analyze the audio;
2. break it into parts;
3. digitize it into a computer-readable format; and
4. use an algorithm to match it to the most suitable text representation.

Speech recognition software must adapt to the highly variable and context-specific nature of human speech. The software algorithms that process and organize audio into text are trained on different speech patterns, speaking styles, languages, dialects, accents and phrasings. The software also separates spoken audio from background noise that often accompanies the signal.

To meet these requirements, speech recognition systems use two types of models:

- **Acoustic models.** These represent the relationship between linguistic units of speech and audio signals.
- **Language models.** Here, sounds are matched with word sequences to distinguish between words that sound similar.

What applications is speech recognition used for?

Speech recognition systems have quite a few applications. Here is a sampling of them.

Mobile devices. Smartphones use voice commands for call routing, speech-to-text processing, voice dialing and voice search. Users can respond to a text without looking at their devices. On Apple iPhones, speech recognition powers the keyboard and Siri, the virtual assistant. Functionality is available in secondary languages, too. Speech recognition can also be found in word processing applications like Microsoft Word, where users can dictate words to be turned into text.

Education. Speech recognition software is used in language instruction. The software hears the user's speech and offers help with pronunciation.

Customer service. Automated voice assistants listen to customer queries and provides helpful resources.

Healthcare applications. Doctors can use speech recognition software to transcribe notes in real time into healthcare records.

Disability assistance. Speech recognition software can translate spoken words into text using closed captions to enable a person with hearing loss to understand what others are saying. Speech recognition can also enable those with limited use of their hands to work with computers, using voice commands instead of typing.

Court reporting. Software can be used to transcribe courtroom proceedings, precluding the need for human transcribers.

Emotion recognition. This technology can analyze certain vocal characteristics to determine what emotion the speaker is feeling. Paired with sentiment analysis, this can reveal how someone feels about a product or service.

Advantages of speech recognition?

There are several advantages to using speech recognition software, including the following:

- **Machine-to-human communication.** The technology enables electronic devices to communicate with humans in natural language or conversational speech.
- **Readily accessible.** This software is frequently installed in computers and mobile devices, making it accessible.
- **Easy to use.** Well-designed software is straightforward to operate and often runs in the background.
- **Continuous, automatic improvement.** Speech recognition systems that incorporate AI become more effective and easier to use over time. As systems complete speech recognition tasks, they generate more data about human speech and get better at what they do.

Disadvantages of speech recognition?

While convenient, speech recognition technology still has a few issues to work through. Limitations include:

- **Inconsistent performance.** The systems may be unable to capture words accurately because of variations in pronunciation, lack of support for some languages and inability to sort through background noise. Ambient noise can be especially challenging. Acoustic training can help filter it out, but these programs aren't perfect. Sometimes it's impossible to isolate the human voice.
- **Speed.** Some speech recognition programs take time to deploy and master. The speech processing may feel relatively slow**.**
- **Source file issues.** Speech recognition success depends on the recording equipment used, not just the software.

What are the different speech recognition algorithms?

The power behind speech recognition features comes from a set of algorithms and technologies. They include the following:

- **Hidden Markov model.** HMMs are used in autonomous systems where a state is partially observable or when all of the information necessary to make a decision is not immediately available to the sensor (in speech recognition's case, a microphone). An example of this is in acoustic modeling, where a program must match linguistic units to audio signals using statistical probability.
- **Natural language processing.** NLP eases and accelerates the speech recognition process.
- **N-grams.** This simple approach to language models creates a probability distribution for a sequence. An example would be an algorithm that looks at the last few words spoken, approximates the history of the sample of speech and uses that to determine the probability of the next word or phrase that will be spoken.
- **Artificial intelligence.** AI and machine learning methods like deep learning and neural networks are common in advanced speech recognition software. These systems use grammar, structure, syntax and composition of audio and voice signals to process speech. Machine learning systems gain knowledge with each use, making them well suited for nuances like accents.

Image processing and computer vision

Computer vision

Computer vision is a field of artificial intelligence (AI) that uses machine learning and neural networks to teach computers and systems to derive meaningful information from digital images, videos and other visual inputs—and to make recommendations or take actions when they see defects or issues.

History of Computer Vision

Computer vision is not a new technology because scientists and experts have been trying to develop machines that can see and understand visual data for almost six decades. The evolution of computer vision is classified as follows:

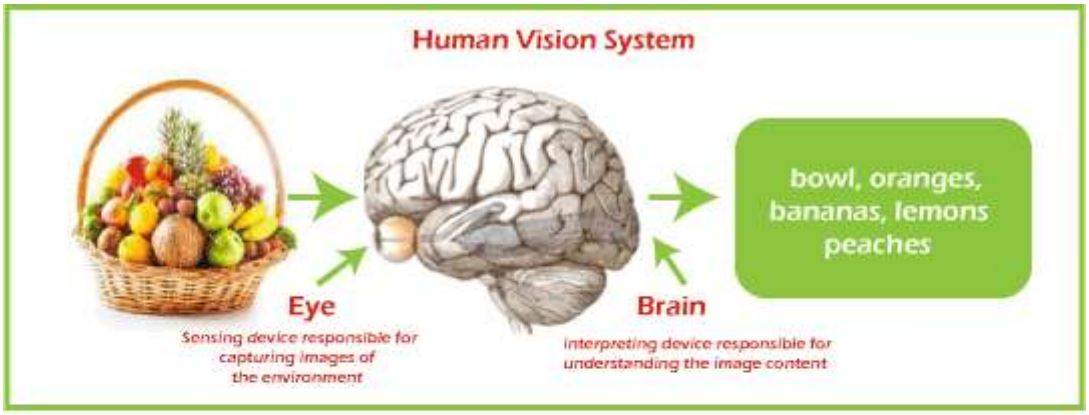
- **1959:** The first experiment with computer vision was initiated in 1959, where they showed a cat as an array of images. Initially, they found that the system reacts first to

hard edges or lines, and scientifically, this means that image processing begins with simple shapes such as straight edges.

- **1960:** In 1960, artificial intelligence was added as a field of academic study to solve human vision problems.
- **1963:** This was another great achievement for scientists when they developed computers that could transform 2D images into 3-D images.
- **1974:** This year, optical character recognition (OCR) and intelligent character recognition (ICR) technologies were successfully discovered. The OCR has solved the problem of recognizing text printed in any font or typeface, whereas ICR can decrypt handwritten text. These inventions are one of the greatest achievements in document and invoice processing, vehicle number plate recognition, mobile payments, machine translation, etc.
- **1982:** In this year, the algorithm was developed to detect edges, corners, curves, and other shapes. Further, scientists also developed a network of cells that could recognize patterns.
- **2000:** In this year, scientists worked on a study of object recognition.
- **2001:** The first real-time face recognition application was developed.
- **2010:** The ImageNet data set became available to use with millions of tagged images, which can be considered the foundation for recent Convolutional Neural Network (CNN) and deep learning models.
- **2012:** CNN has been used as an image recognition technology with a reduced error rate.
- **2014:** COCO has also been developed to offer a dataset for object detection and support future research.

How does Computer Vision Work?

Computer vision is a technique that extracts information from visual data, such as images and videos. Although computer vision works similarly to human eyes with brain work, this is probably one of the biggest open questions for IT professionals: How does the human brain operate and solve visual object recognition?

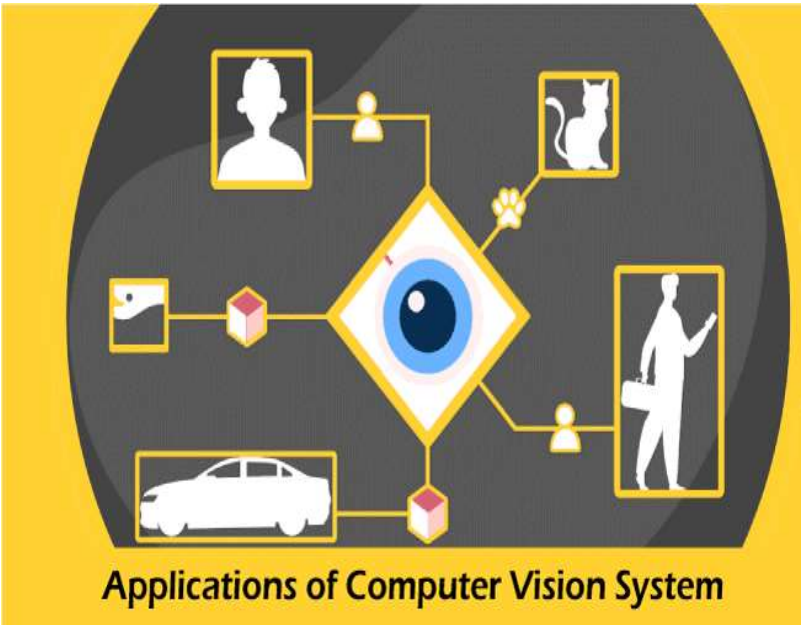


On a certain level, computer vision is all about pattern recognition which includes the training process of machine systems for understanding the visual data such as images and videos, etc.

Firstly, a vast amount of visual labeled data is provided to machines to train it. This labeled data enables the machine to analyze different patterns in all the data points and can relate to those labels. E.g., suppose we provide visual data of millions of dog images. In that case, the computer learns from this data, analyzes each photo, shape, the distance between each shape, color, etc., and hence identifies patterns similar to dogs and generates a model. As a result, this computer vision model can now accurately detect whether the image contains a dog or not for each input image.

Applications of computer vision

Computer vision is one of the most advanced innovations of artificial intelligence and machine learning. As per the increasing demand for AI and Machine Learning technologies, computer vision has also become a center of attraction among different sectors. It greatly impacts different industries, including retail, security, healthcare, automotive, agriculture, etc.



Below are some most popular applications of computer vision:

- **Facial recognition:** Computer vision has enabled machines to detect face images of people to verify their identity. Initially, the machines are given input data images in which computer vision algorithms detect facial features and compare them with databases of fake profiles. Popular social media platforms like Facebook also use facial recognition to detect and tag users. Further, various government spy agencies are employing this feature to identify criminals in video feeds.
- **Healthcare and Medicine:** Computer vision has played an important role in the healthcare and medicine industry. Traditional approaches for evaluating cancerous tumors are time-consuming and have less accurate predictions, whereas computer vision technology provides faster and more accurate chemotherapy response assessments; doctors can identify cancer patients who need faster surgery with life-saving precision.
- **Self-driving vehicles:** Computer vision technology has also contributed to its role in self-driving vehicles to make sense of their surroundings by capturing video from different angles around the car and then introducing it into the software. This helps to detect other cars and objects, read traffic signals, pedestrian paths, etc., and safely drive its passengers to their destination.
- **Optical character recognition (OCR)**
  Optical character recognition helps us extract printed or handwritten text from visual data such as images. Further, it also enables us to extract text from documents like invoices, bills, articles, etc.

- **Machine inspection:** Computer vision is vital in providing an image-based automatic inspection. It detects a machine's defects, features, and functional flaws, determines inspection goals, chooses lighting and material-handling techniques, and other irregularities in manufactured products.
- **Retail (e.g., automated checkouts):** Computer vision is also being implemented in the retail industries to track products, shelves, wages, record product movements into the store, etc. This AI-based computer vision technique automatically charges the customer for the marked products upon checkout from the retail stores.
- **3D model building:** 3D model building or 3D modeling is a technique to generate a 3D digital representation of any object or surface using the software. In this field also, computer vision plays its role in constructing 3D computer models from existing objects. Furthermore, 3D modeling has a variety of applications in various places, such as Robotics, Autonomous driving, 3D tracking, 3D scene reconstruction, and AR/VR.
- **Medical imaging:** Computer vision helps medical professionals make better decisions regarding treating patients by developing visualization of specific body parts such as organs and tissues. It helps them get more accurate diagnoses and a better patient care system. E.g., Computed Tomography (CT) or Magnetic Resonance Imaging (MRI) scanner to diagnose pathologies or guide medical interventions such as surgical planning or for research purposes.
- **Automotive safety:** Computer vision has added an important safety feature in automotive industries. E.g., if a vehicle is taught to detect objects and dangers, it could prevent an accident and save thousands of lives and property.
- **Surveillance:** It is one of computer vision technology's most important and beneficial use cases. Nowadays, CCTV cameras are almost fitted in every place, such as streets, roads, highways, shops, stores, etc., to spot various doubtful or criminal activities. It helps provide live footage of public places to identify suspicious behavior, identify dangerous objects, and prevent crimes by maintaining law and order.
- **Fingerprint recognition and biometrics:** Computer vision technology detects fingerprints and biometrics to validate a user's identity. Biometrics deals with recognizing persons based on physiological characteristics, such as the face, fingerprint, vascular pattern, or iris, and behavioral traits, such as gait or speech. It combines Computer Vision with knowledge of human physiology and behavior.

Image processing

Image processing is the process of transforming an image into a digital form and performing certain operations to get some useful information from it. The image processing system usually treats all images as 2D signals when applying certain predetermined signal processing methods.

Types of Image Processing

There are five main types of image processing:

- Visualization - Find objects that are not visible in the image
- Recognition - Distinguish or detect objects in the image
- Sharpening and restoration - Create an enhanced image from the original image
- Pattern recognition - Measure the various patterns around the objects in the image
- Retrieval - Browse and search images from a large database of digital images that are similar to the original image

Components of Image Processing

Computer

A general-purpose computer, which may be anything from a PC to a supercomputer, is used in an image processing system. Sometimes, specifically built computers are utilized in specialized applications to reach a specified degree of performance.

Hardware for Specialized Image Processing

It comprises the digitizer and hardware that can carry out basic operations, including an Arithmetic Logic Unit (ALU), which can carry out simultaneous arithmetic and logical operations on whole pictures.

Massive Storing

In applications involving image processing, the skill is essential. The three main types of digital storage for image processing applications are as follows: Three types of storage exist (1) short-term storage, (2) online storage for quick recall (3) archive storage, which is characterized by rare access.

Camera Sensors

It alludes to perception. The image sensor's primary function is to collect incoming light, transform it into an electrical signal, measure that signal, and then output it to supporting electronics. It consists of a two-dimensional array of light-sensitive components that convert photons into electrons. Images are captured by equipment like digital cameras using image sensors like CCD and CMOS. Two components are often needed on image sensors to collect digital pictures. The first is an actual tool (sensor) that can detect the energy emitted by the object we want to turn into an image. The second is a digitizer, which transforms a physical sensing device's output into digital form.

Image Display

The pictures are shown.

Software

The image processing software comprises specialized modules that carry out particular functions.

Hardcopy Equipment

Laser printers, film cameras, heat-sensitive equipment, inkjet printers, and digital equipment like optical and CDROM discs are just a few examples of the instruments used to record pictures.

Networking

To send visual data through a networked computer, it is a necessary component. The most important factor in picture transmission is bandwidth since image processing applications require vast amounts of data.

Robotics

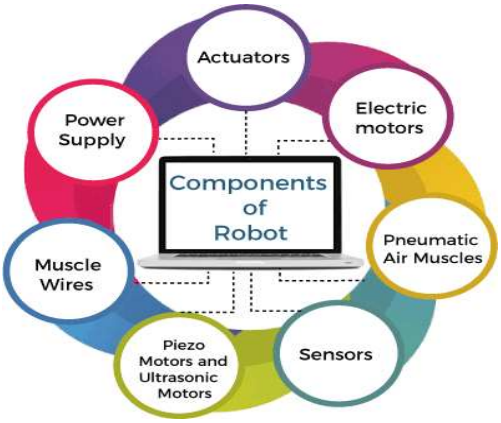
Robotics is a separate entity in Artificial Intelligence that helps study the creation of intelligent robots or machines. Robotics combines electrical engineering, mechanical engineering and computer science & engineering as they have mechanical construction, electrical component and programmed with programming language. Although, Robotics and Artificial Intelligence both have different objectives and applications, but most people treat robotics as a subset of Artificial Intelligence (AI). Robot machines look very similar to humans, and also, they can perform like humans, if enabled with AI.

What is a robot?

A robot is a machine that looks like a human, and is capable of performing out of box actions and replicating certain human movements automatically by means of commands given to it using programming. Examples: Drug Compounding Robot, Automotive Industry Robots, Order Picking Robots, Industrial Floor Scrubbers and Sage Automation Gantry Robots, etc.

Components of Robot

Several components construct a robot, these components are as follows:



- **Actuators:** Actuators are the devices that are responsible for moving and controlling a system or machine. It helps to achieve physical movements by converting energy like electrical, hydraulic and air, etc. Actuators can create linear as well as rotary motion.
- **Power Supply:** It is an electrical device that supplies electrical power to an electrical load. The primary function of the power supply is to convert electrical current to power the load.
- **Electric Motors:** These are the devices that convert electrical energy into mechanical energy and are required for the rotational motion of the machines.
- **Pneumatic Air Muscles:** Air Muscles are soft pneumatic devices that are ideally best fitted for robotics. They can contract and extend and operate by pressurized air filling a pneumatic bladder. Whenever air is introduced, it can contract up to 40%.
- **Muscles wire:** These are made up of nickel-titanium alloy called Nitinol and are very thin in shape. It can also extend and contract when a specific amount of heat and electric current is supplied into it. Also, it can be formed and bent into different shapes when it is in its martensitic form. They can contract by 5% when electrical current passes through them.
- **Piezo Motors and Ultrasonic Motors:** Piezoelectric motors or Piezo motors are the electrical devices that receive an electric signal and apply a directional force to an opposing ceramic plate. It helps a robot to move in the desired direction. These are the best suited electrical motors for industrial robots.
- **Sensor:** They provide the ability like see, hear, touch and movement like humans. *Sensors are the devices or machines which help to detect the events or changes in the environment and send data to the computer processor. These devices are usually equipped with other electronic devices*. Similar to human organs, the electrical sensor also plays a crucial role in Artificial Intelligence & robotics. AI

algorithms control robots by sensing the environment, and it provides real-time information to computer processors.

Applications of Robotics

Robotics have different application areas. Some of the important applications domains of robotics are as follows:

- **Robotics in defence sectors:** The defence sector is undoubtedly the one of the main parts of any country. Each country wants their defence system to be strong. Robots help to approach inaccessible and dangerous zone during war. DRDO has developed a robot named ***Daksh*** to destroy life-threatening objects safely. They help soldiers to remain safe and deployed by the military in combat scenarios. Besides combat support, robots are also deployed in ***anti-submarine operations, fire support, battle damage management, strike missions, and laying machines.***
- **Robotics in Medical sectors:** Robots also help in various medical fields such as laparoscopy, neurosurgery, orthopaedic surgery, disinfecting rooms, dispensing medication, and various other medical domains.
- **Robotics in Industrial Sector:** Robots are used in various industrial manufacturing industries such as cutting, welding, assembly, disassembly, pick and place for printed circuit boards, packaging & labelling, palletizing, product inspection & testing, colour coating, drilling, polishing and handling the materials. Moreover, Robotics technology increases productivity and profitability and reduces human efforts, resulting from lower physical strain and injury. The industrial robot has some important advantages, which are as follows:
    - Accuracy
    - Flexibility
    - Reduced labour charge
    - Low noise operation
    - Fewer production damages
    - Increased productivity rate.
- **Robotics in Entertainment:** Over the last decade, use of robots is continuously getting increased in entertainment areas. Robots are being employed in entertainment sector, such as movies, animation, games and cartoons. Robots are very helpful where repetitive actions are required. A camera-wielding robot helps shoot a movie scene as

many times as needed without getting tired and frustrated. A big-name **Disney** has launched hundreds of robots for the film industry.

- **Robots in the mining industry:** Robotics is very helpful for various mining applications such as robotic dozing, excavation and haulage, robotic mapping & surveying, robotic drilling and explosive handling, etc. A mining robot can solely navigate flooded passages and use cameras and other sensors to detect valuable minerals. Further, robots also help in excavation to detect gases and other materials and keep humans safe from harm and injuries. The robot rock climbers are used for space exploration, and underwater drones are used for ocean exploration.